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CHAPTER 17

Action–perception connection and the cortical mu rhythm

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Abstract: The rolandic mu rhythm consists of two main frequency components: one around 10 Hz and the other around 20 Hz. Reactivity of the mu rhythm, especially its motor cortex 20-Hz component, provides an illuminating window to the involvement of the human sensorimotor system in the loop that connects action and perception with the environment.

Keywords: MEG; EEG; motor cortex; cortical oscillations; mirror-neuron system; action–perception link

Introduction

In their insightful publication "EEG changes during cinematographic presentation," Gastaut and Bert (1954) noted half a century ago an important association between reactivity of rhythmic activity in the rolandic region and the motor attitudes of persons watching a movie on a boxing match. According to the authors, the "rhythms en arceau" were blocked whenever spectators identified themselves with one of the active figures projected on the screen. Along similar lines, this chapter discusses the relationship between the action, perception, and the human rolandic oscillations, with a strong emphasis on the 20-Hz motor–cortex activity.

I prefer to define magnetoencephalographic (MEG) and electroencephalographic (EEG) rhythms according to their (approximate) cerebral origin and their typical frequency, avoiding terms synchronization and desynchronization. This is because the disappearance (appearance) of an MEG/EEG rhythm can be either due to desynchronization (synchronization) of the underlying continuously functioning oscillators or due to a standstill (wake up) of the generator population. On the basis of MEG and EEG recordings, we just cannot tell.

As an important addition to scalp EEG recordings, MEG allows the same brain currents to be seen "directly through the skull" because tissues around the brain do not significantly distort the magnetic fields (Hämäläinen, et al., 1993; Hari et al., 2000). The most synchronous activation dominates both MEG and EEG signals.

The modern neuromagnetometers allow the whole brain's magnetic field patterns to be recorded at once and they thereby provide powerful tools for both basic and applied human neuroscience. MEG is most sensitive to activation of fissural cortex where currents flow tangential to the skull; these areas are difficult to reach with other means, including intracranial EEG recordings. In the rolandic region, the histologically defined motor cortex (area 4) is buried within the central sulcus from the most lateral face areas up to the more medial hand region. Thus MEG's sensitivity to activity of fissural cortex is an advantage in picking up rolandic mu rhythm that concentrates on the hand region.

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Rolandic mu rhythm

Because of its arch-like appearance, the rolandic mu rhythm cannot consist of one frequency only but comprises (at least) two spectral peaks, one around 10 Hz and the other around 20 Hz; typically these frequencies are not exact harmonics (Tiihonen et al., 1989). The levels of both frequencies, studied after bandpass filtering, co-vary to a great extent in association with voluntary movements but with striking differences: the 10-Hz activity starts to suppress earlier than the 20-Hz activity before the movement, whereas the 20-Hz level shows a more clear and earlier "rebound" after the movement (Pfurtscheller, 1981; Salmelin and Hari, 1994; Nagamine et al., 1996). Whereas the post movement ~20-Hz rebounds follow in a somatotopic manner the moving body part (appearing lateral in the rolandic cortex after mouth movements, more medial after finger movements, and close to midline after foot movements), the ~10-Hz activity tends to cluster close to the hand region (Salmelin et al., 1995).

The 10-Hz rolandic MEG activity is dampened by motor activity or tactile stimulation, as well as during binocular rivalry when the visual percept changes without any modification of the incoming information (Vanni et al., 1999). This reactivity seems ecologically valid in assuring that the sensorimotor cortex is automatically informed when visual guidance of movements would be needed. The reactivity of both 10- and 20-Hz frequency components in association with even unilateral movements is typically bilateral, although with contralateral dominance (Salenius et al., 1997; Hari et al., 1998).

Figure 1 shows rebounds in the 20-Hz activity after brisk lifting movements of the right index finger; 13 bandpass filtered MEG traces from the left rolandic cortex are superimposed. Very similar reactivity also occurs after electric median nerve stimuli: the ~20-Hz rhythm is first transiently, and bilaterally, suppressed and then strongly enhanced, with a peak at 500–700 ms (Salmelin and Hari, 1994; see also Fig. 2).

As discussed in recent review papers (Hari and Salmelin, 1997; Hari and Salenius, 1999; Salenius and Hari, 2003), several lines of evidence suggest that the ~20-Hz MEG and EEG rhythms originate predominantly in the precentral primary motor cortex (Hari and Salmelin, 1997). The main arguments are the following:

1. *Intracranial recordings*. Oscillatory activity of about 25 Hz has been recorded directly from the human primary motor cortex, from the anterior wall of the central sulcus (Jasper and Penfield, 1949).
2. *Source analysis*. Current dipole sources of MEG rolandic rhythms cluster more anteriorly for the ~20-Hz component than for the 10-Hz component, suggesting predominant contributions from the primary motor and primary somatosensory cortices in the anterior and posterior walls of the central sulcus, respectively (Salmelin and Hari, 1994).

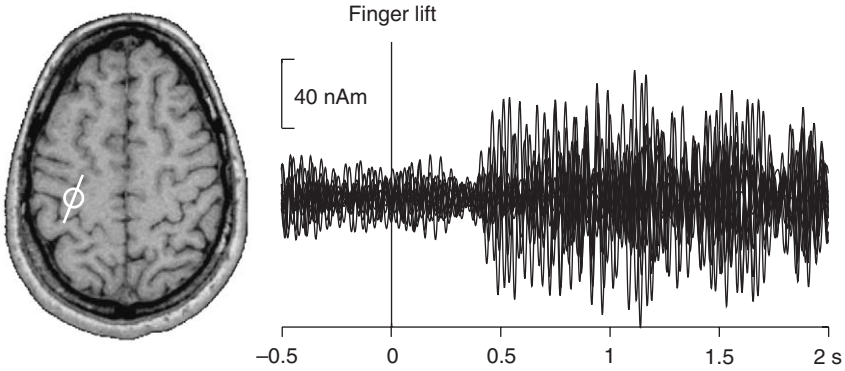

Fig. 1. Reactivity of the motor–cortex 20-Hz activity during 13 separate finger lifts occurring at time 0. The circle on the brain indicates the site of the source cluster for these oscillations; the line shows the main orientation of the current dipoles. Adapted from Longcamp et al. (2006).

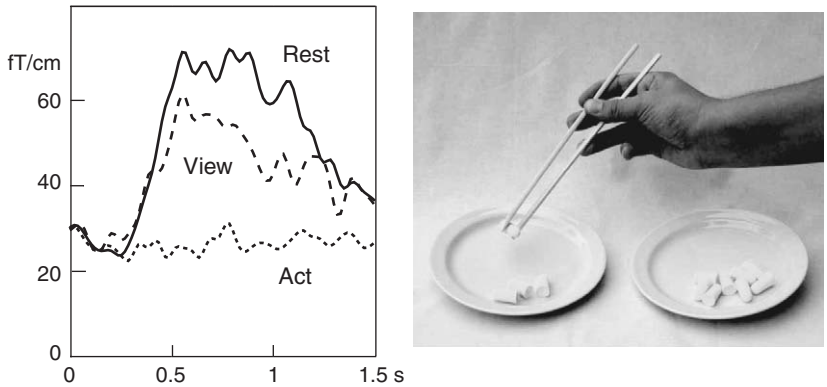

Fig. 2. Rebounds of motor–cortex 20-Hz oscillations after median-nerve stimulation during rest, during own finger movements (act), and during observation of chopstick use by another person. Modified from Järveläinen et al. (2004).

3. *Cortex–muscle coherence.* During isometric muscle contraction, the motor cortex activity is coherent with the motor unit firing at ~20 Hz during intermediate force (Conway et al., 1995; Salenius et al., 1997) or at 40 Hz during strong force (Brown et al., 1998). Cortex leads in time, with time lags compatible with conduction times from motor cortex to muscles (Salenius et al., 1997). Importantly, the sites of the strongest coherence agree with the cortical control sites of the muscles involved, and the motor cortex origin of the coherent activity has been confirmed by intraoperative electrical stimulation (Mäkelä et al., 2001) and, more recently, by studies of patients with early-age lesions of the motor cortex (Gerloff et al., 2006).
4. *Effects of benzodiazepine*. Administration of GABA-ergic benzodiazepine increases 20-Hz activity in the primary motor cortex close to the hand area (Jensen et al., 2005).

A recent intracranial study on three epileptic patients challenged the dominance of the motor cortex in the generation of the rolandic 20-Hz activity (Szurhaj et al., 2003). However, even the intracranial recordings are prone to localization errors caused by volume conduction, direction of the source currents, and location of the reference electrode.

Cortical oscillations and functional state of motor cortex

The level of the 20-Hz activity after movements or median-nerve stimulation is a reliable tool to monitor the functional state of the primary motor cortex, e.g., to indicate that acute pain automatically excites the cortex (Raij et al., 2004). A transcranial magnetic stimulation study demonstrated decreased cortical excitability after median nerve stimulation, with a similar time course as observed for the 20-Hz rhythm rebound (Chen et al., 1999).

These data agree with the early view that the human mu rhythm, especially the ~20-Hz activity, is associated with the subject's immobility and therefore most likely with cortical inhibition (Jasper and Penfield, 1949; Gastaut, 1952). The enhancement of the 20-Hz activity by oral administration of GABA-ergic benzodiazepine further supports this interpretation, as does neural network modeling that demonstrates the importance of inhibitory postsynaptic currents for the generation of the 20-Hz rhythm (Jensen et al., 2005).

On the other hand, a relationship between cortical inhibition and motor system stabilization is suggested by findings that direct application of GABA-A antagonist into motor cortex of monkey results in clumsy movements and spontaneous muscle twitches (Matsumura et al., 1991). Along similar lines, studies of task-dependent modulation of cortex muscle coherence during a variety of precision grip tasks suggest that the 20-Hz rebound

is related to resetting of the control system involving the motor cortex after movements (Kilner et al., 2000). Accordingly, our current working hypothesis is that the high level of the mu rhythm is related to stabilization of the motor system. Further indicators of the relationship between the excitation/inhibition balance and the motor cortex rhythms are seen in patients with chronic pain or progressive myoclonus epilepsy who have deficient 20-Hz rebounds as indicators of cortical disinhibition (Silén et al., 2000; Juottonen et al., 2002).

Motor cortex and mirror neurons

The 20-Hz rebound after median nerve stimulation is totally abolished when the subject herself moves the fingers or manipulates a small object, indicating clear activation of the primary motor cortex. Interestingly, a similar—although significantly weaker—activation also occurs when the subject just views another person's manipulation movements without moving herself (Hari et al., 1998). This suppression occurs despite totally silent surface eletromyogram, confirming that the subject did not herself contract muscles while viewing the other person's movements. The suppression of the rebound is 15–20% stronger for motor acts presented live than those seen on a video (Järveläinen et al., 2001).

MEG findings about activation of the viewer's motor cortex, also well evident in EEG recordings and in transcranial magnetic stimulation studies (for reviews, see Fadiga et al., 2005; Pineda, 2005), can be interpreted in the framework of mirror neurons. In their pioneering studies on monkey, the Parma neuroscientists (Gallese et al., 1996; Rizzolatti et al., 1996) noted that area F5 in monkey ventral premotor cortex contains neurons that discharge when monkeys perform goal-directed hand movements and when they observe another monkey or human execute similar movements. Since then these "mirror neurons" have been suggested to directly match observed and executed actions. The human counterpart for the monkey F5 area is the inferior frontal gyrus, IFG (called Broca's region in the left hemisphere). Functional neuroimaging and electrophysiological studies indicate this region to behave like it would contain motor mirror neurons (for a review, see Rizzolatti and Craighero, 2004).

The human mirror-neuron system (MNS) includes, in addition to the IFG, the primary motor cortex, which—importantly for the present discussions—is downstream from the IFG, as is clearly indicated by activation latencies (Nishitani and Hari, 2000, 2002). Therefore, the motor cortex reactivity easily reflects function of the IFG, the core of the human MNS. The IFG, possibly with the inferior parietal lobe, could serve as an interface for action–perception connections and for internal forward and backward models (for a review, see Nishitani et al., 2005); the forward models inform, via efference copies, the sensory areas about the forthcoming sensory input, which has to be compared with the predictions. Moreover, MEG timing sequences suggest that the human superior temporal sulcus is connected to the IFG via the inferior parietal lobule (Nishitani and Hari, 2002). The MNS provides a mechanism for shared motor representations, and it is closely interconnected to other brain areas that likely support sensory mirroring of e.g., touch (Avikainen et al., 2002).

Whereas the monkey mirror neurons react only to direct hand–object contact, the human MNS reacts also to tool use, which humans practice throughout their life span. When healthy subjects observed the experimenter to use chopsticks, the 20-Hz motor-cortex rebounds elicited by median nerve stimulation were suppressed (Fig. 2), and the suppression was 15–17% stronger for goal-directed than non-goal-directed chopsticks movements (Järveläinen et al., 2004). Interestingly, this difference was positively correlated with the frequency of subjects' chopstick use during the last year. Thus, the motor-cortex activation seems to be related to the observer's own motor skills and thereby to the ability to understand and imitate motor acts.

Covariation of the actor's and observer's motor systems

Observation of other persons' actions seems to lead in automatized action simulation and embodied perception. The reactions are predictive, as has been demonstrated by eye movement recordings

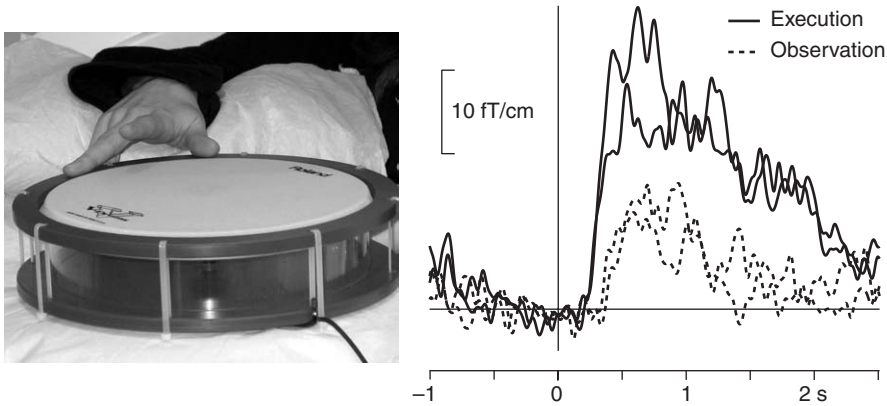

Fig. 3. Movement-related rebounds of the motor–cortex 20-Hz activity when the subject was either tapping a drum himself (execution) or observing another person's similar movements (observation). Traces from two similar conditions are superimposed. Please note that these envelopes illustrate phenomena similar to those in the single traces of Fig. 1, whereas the rebounds in Fig. 2 were elicited by median-nerve stimuli. Adapted from Caetano et al. (2006).

(Flanagan and Johansson, 2003) and by premovement EEG shifts in the observer's brain (Kilner et al., 2004). The system of shared motor representations, with Broca's region as the core (for reviews, see Rizzolatti and Craighero, 2004; Nishitani et al., 2005), is important for understanding intentions and goals of other persons, contributing to the brain basis of intersubjectivity.

The similarities between the first and second/third person motor actions are further supported by findings presented in Fig. 3 (Caetano et al., 2006): When a subject taps a drum membrane with the right index finger, a typical 20-Hz rebound appears in the contralateral motor cortex. However, a similar but considerably smaller rebound, delayed by about 115 ms, is also seen when the person just observes another person tapping the drum. Altogether the available evidence, including action–observation-related EEG data (Babiloni et al., 2002), indicates highly similar neuronal mechanisms underlying first person performance and second/third person observation of motor actions: motor cortex is activated before and during the observed motor actions, and stabilized afterwards.

Action–perception loop and the mu rhythm

Common to all successful communication is that the sender and the receiver of a message stay tuned. For example, the motor theory of speech (Liberman and Whalen, 2000) proposes that the listeners match their own motor representations to the motor articulation patterns of the speaker. Furthermore, during social interaction in general, one individual's output is the other (similar) individual's input. This leads to action–perception correspondence *within* and *between* individuals. The between-individuals matching likely involves the motor MNS, reflected in the motor cortex reactivity already in early childhood (Fecteau et al., 2004).

Interestingly, highly functioning autistic Asperger subjects, who are impaired in imitation and in reading other persons' body language, show both dampening and delay of responses in the inferior frontal cortex when they have to imitate orofacial gestures (Nishitani et al., 2004). Downstream from the IFG, the primary motor cortex of Asperger persons reacts to observed movements of other persons, although with a trend to (Avikainen et al., 1999) or with clear signs of (Oberman et al., 2005) decreased reactivity. Such impaired reactivity, however, has to be interpreted with caution because of difficulties in controlling the autistic person's attention to human stimuli. Instead, the observed motor cortex reactivity is a reliable finding, clearly ruling out a totally non-functional MNS in Asperger subjects.

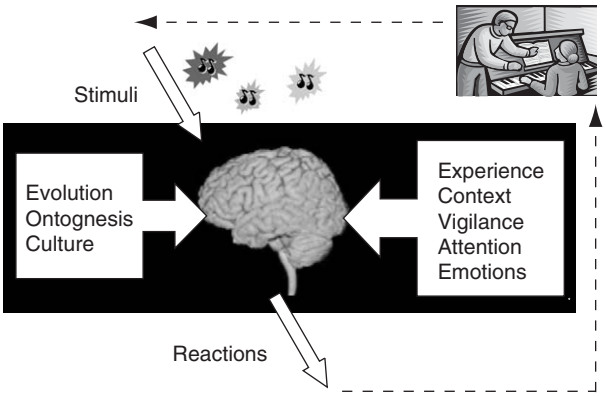

Fig. 4. A schematic presentation of the action–perception loop: The reactions in any situation are shaped by evolution, ontogenesis, and culture of the individual brain, and by the currently available experience (memory), context, vigilance, attention, and emotions. Importantly, the action–perception loop is closed via the environment and thereby includes active interaction with both inanimate and animate world.

Figure 4 illustrates schematically the relationship between action and perception that are closely linked and form a loop that closes via the environment: During tool use or e.g., piano playing, we shape our environment by our own actions and our environment shapes and stimulates us. Similar brain phenomena occur during both overt and covert behaviors and they are reflected in the modulation of the mu rhythm. Repeated cycles of the action–perception loop lead in simulated behavior, comprising predictions of both perceptual and action aspects of the behavior, so that finally a vivid inner world emerges (Jeannerod, 2001; Hesslow, 2002).

The mu rhythm was in the laboratory jargon of 1950s called "the rhythm of subjects ill at ease in their skin" (Gastaut and Bert, 1954). At that time, the rhythm was observed mainly in subjects suffering from some psychopathology, whereas the present evidence, with better source models, indicates that the mu rhythm is a part of normal physiology. The prominence of mu rhythm in subjects with some psychopathological traits could, however, be related to the involvement of the motor system in mental symptoms, in full agreement with the modern considerations of motor cognition (Jeannerod, 2001).

The reactivity of the rolandic cortical rhythms provides an easily accessible window to the interesting connections between perception and action, and to the subject's relationship to the environment, including other people.

Abbreviations

EEG	electroencephalography
IFG	inferior frontal gyrus
MEG	magnetoencephalography
MNS	mirror-neuron system

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